Universität Zürich[UZH]

Departments of Geography and Chemistry

DSA101.1
Introduction to Data Analytics and Python

Prof. Alexandria Deliz Liang

Prof. Meredith C. Schuman, meredithchristine.schuman@uzh.ch

Who am I?

Meredith C. Schuman (Merry)

Born in Minnesota, USA

2012 PhD in Chemical Ecology, FSU, Jena, DE

2012-2019 Group Leader, Max Planck Institute for Chemical Ecology, Jena, DE

2019-2026 Assistant Professor
Since 2026 Associate Professor for Spatial Genetics of Ecosystems, UZH, CH

MAX-PLANCK-GESELLSCHAFT
Max Planck Institute for Chemical Ecology

University of Zurich[UZH]
Departments of Geography and Chemistry

We want to know why living beings vary and where their variation matters for ecology and for people.

SPATIAL GENETICS

Who are you?

Motivation

Clear communication is key!
In code, precise specifications are possible to avoid this and also for problems being tracable





A Boeing Delta II 7425 expendable launch vehicle lifts off with NASA's Mars Climate Orbiter on Dec. 11, 1998
https://www.simscale.com/blog/nasa-mars-climate-orbiter-metric/

Motivation

What happened: Original authors tried to reproduce it but couldnt get from the raw to the source data and retrace
Some workflow used that could not be reproduced and nobody new, where the raw data came from and likely, raw data was published

If you cant go from your data to your conclusions, its not science

The retracted paper analyzed excitatory synapse number and neurotransmitter release in mouse hippocampal neurons missing a presynaptic cell-adhesion molecule called neurexin-2. Südhof, professor of molecular and cellular physiology and neurosurgery at Stanford University, received the Nobel Prize for some of his previous work on neurexin proteins.

The authors retracted the study "because re-analysis of the original raw data for Figs. 2, 4 and 6 revealed that, although our analyses of the original data are supportive of the conclusions of the paper, unresolvable differences exist between these raw data and the published data source file that cannot be corrected by a simple erratum," according to the 5 March retraction notice.

Raw data = Original data
For Raw --> Source: Cleaning, transformation, conversion
Source data = Direct input data for the analysis

You would publish the source data, not the raw data because:
1) Source data meet the FAIR data and uniform for anyone to read and use
2) Raw data are big

https://www.thetransmitter.org/retraction/nobel-prize-winner-thomas-sudhof-retracts-study/

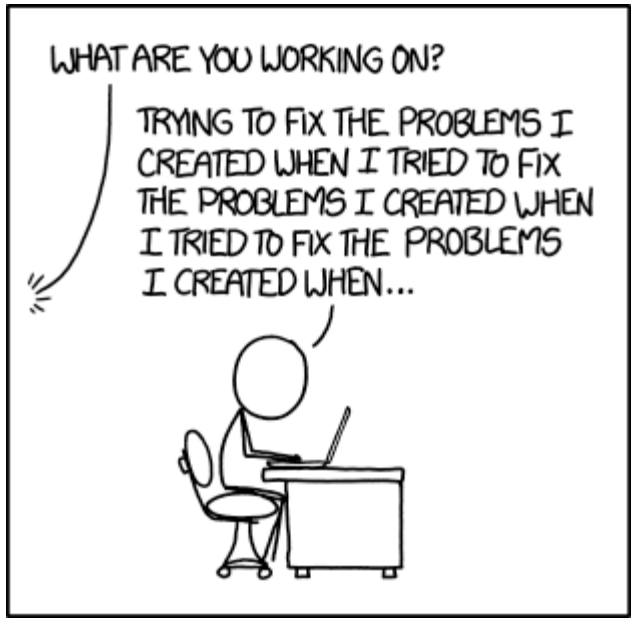
We want to solve a problem using a computer.

There will be many similar, but not identical problems.

Did we solve the original problem correctly?

What if… Did I produce a solution that would still worked if:

- The data change?
- The assumptions change?
- We have 10x as much data? (Or 0.1x as much data?)
- We pass this on to a colleague, or to ourselves six months or six years from now?

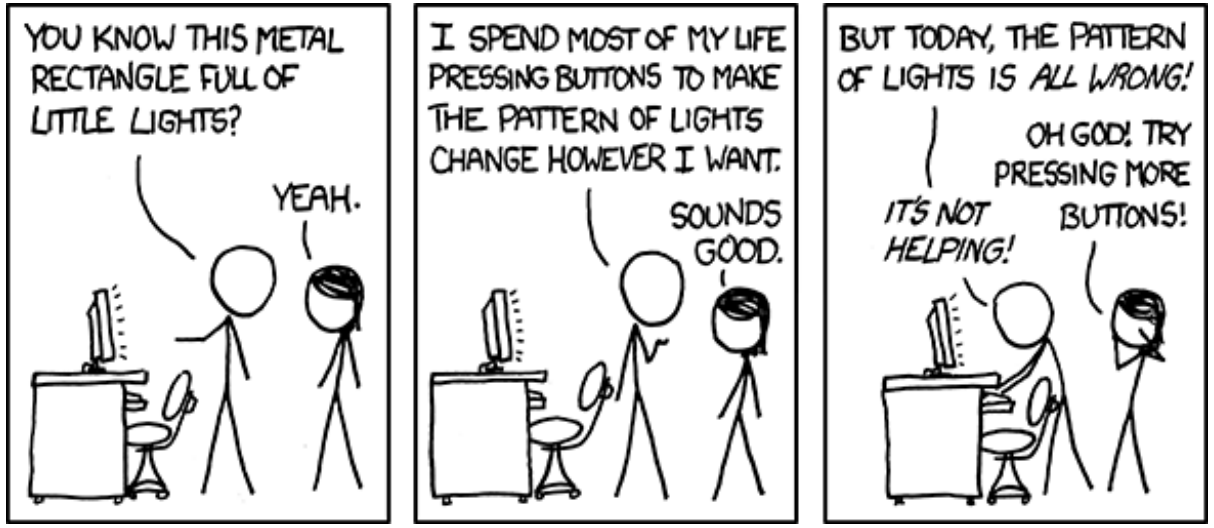


https://www.xkcd.com/1739/

We want to solve a problem using a computer.

If we made a mistake, what happens?

- Program crashes?
- Everything is *increeeedibly* slow?
- It still works for *this specific case*?
- The output is a *little bit* wrong?
- The output is wrong *sometimes*?



https://xkcd.com/722/

We want to solve a problem using a computer.

If we made a mistake, what happens?

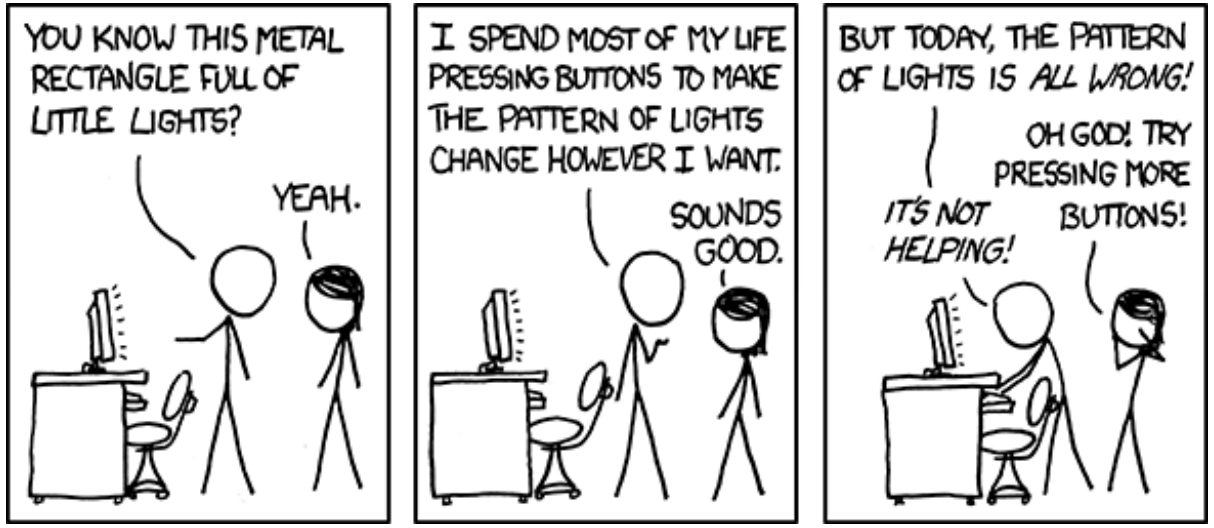
Clear data science ranking:

- Program crashes?
- Everything is *increeeedibly* slow?
- It still works for *this specific case*?
- The output is a *little bit* wrong?
- The output is wrong *sometimes*?

OK

DANGER

What is the worst possible of these



https://xkcd.com/722/



https://xkcd.com/1742/

What is Git?

Git is a program for version control.

What is version control and why is it important?

- You change a file → change can be tracked
- A collaborator changes a file → changes can be merged
- You can go back and forth in time of your project
- You can branch out, test something and either integrate or revert changes
- **You can hardly break anything**

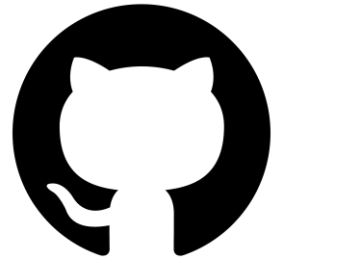


The program
git logo by Jason Long, CC BY 3.0



GitLab

Institutional accounts,
European interface

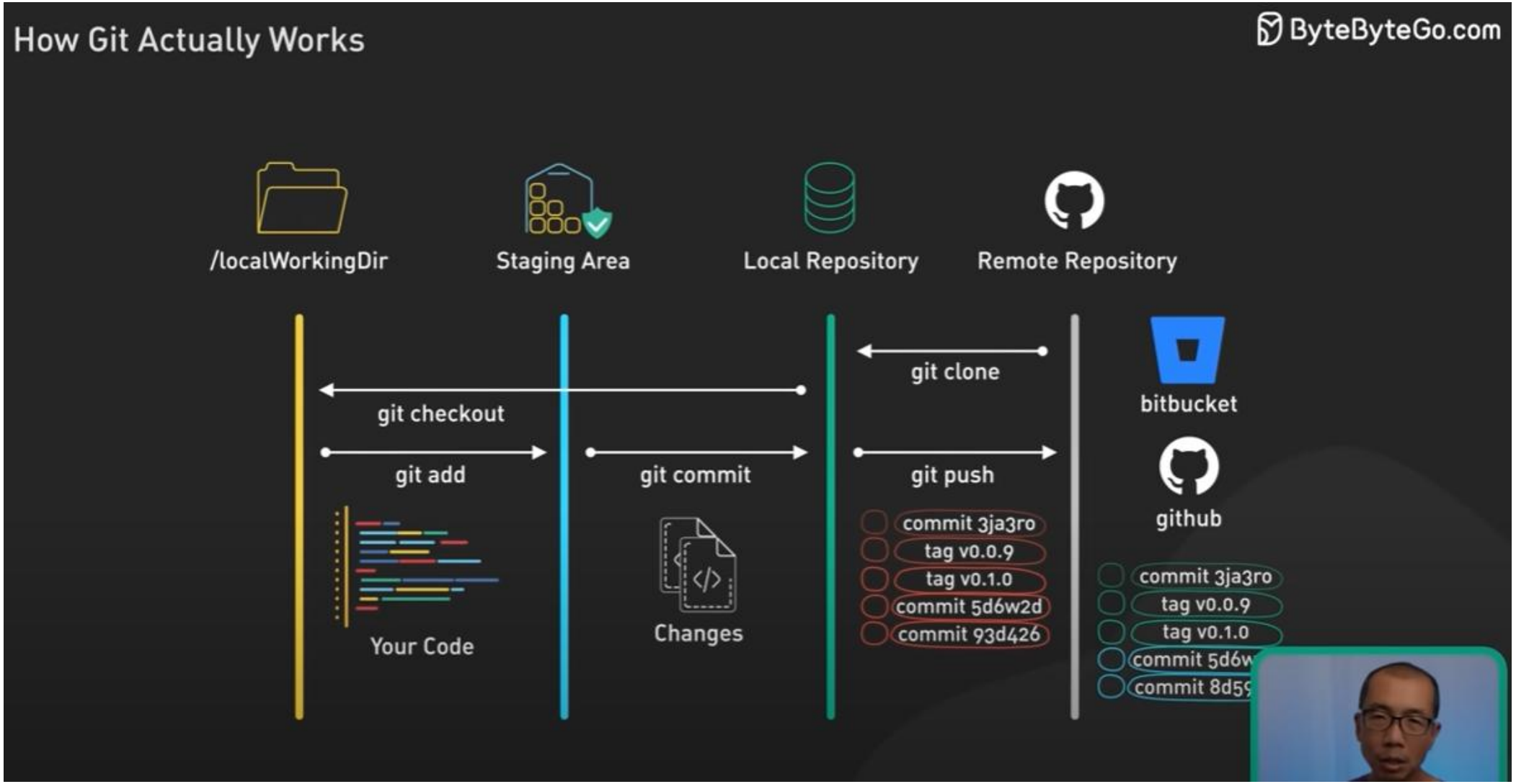


GitHub

Free individual accounts,
North American interface
used worldwide

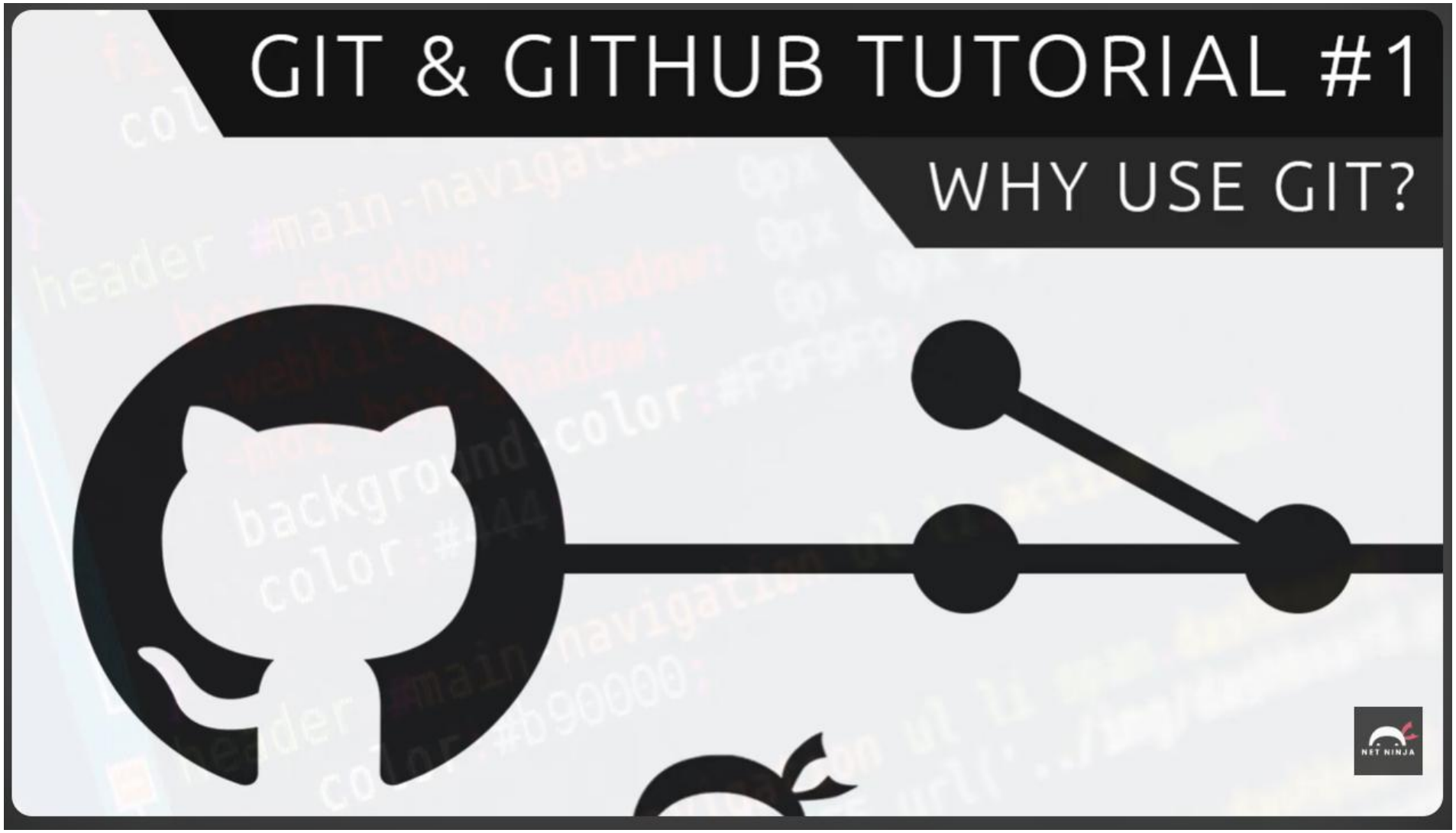
https://docs.gitlab.com/ee/topics/git/

How git and GitHub work



https://youtu.be/e9lnsKot_SQ?si=g0d_pQD2eUPpP8Ae

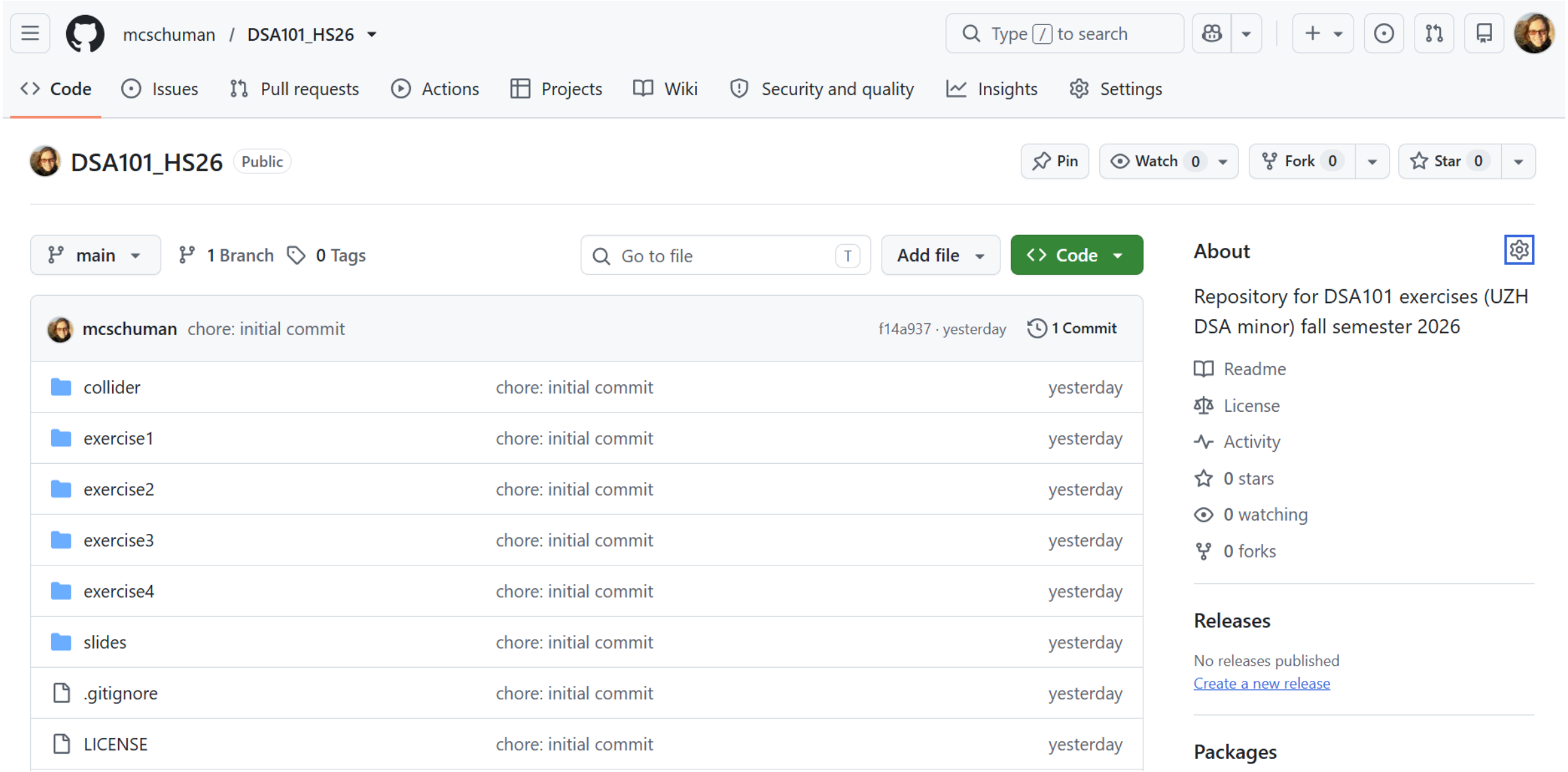
How to git and GitHub



Please watch these GitHub tutorials before we next meet!

You may also find it helpful to refer to the Git Cheat Sheet

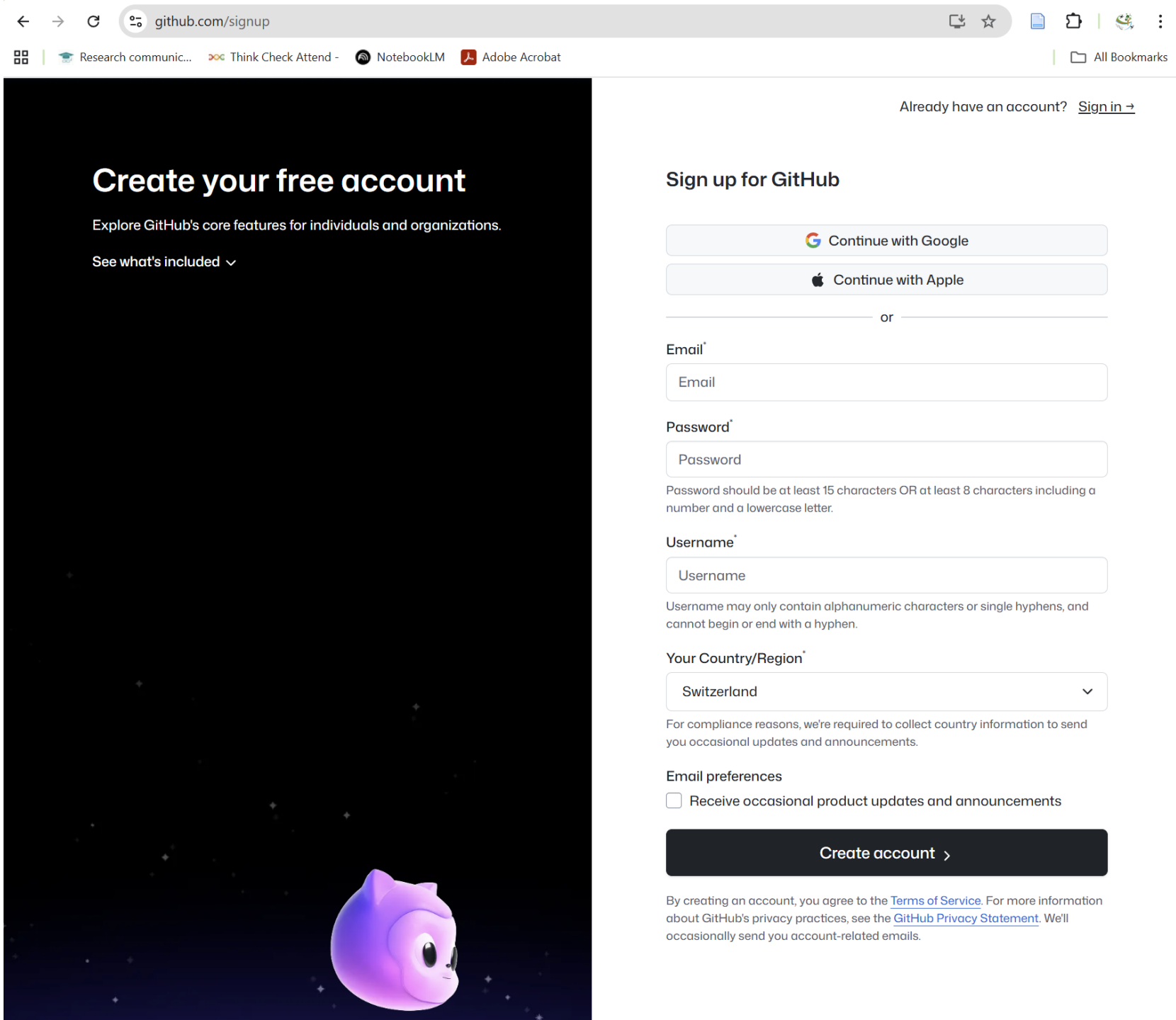
We will be working from a GitHub repo



https://github.com/mcschuman/DSA101_HS26

Let’s start!

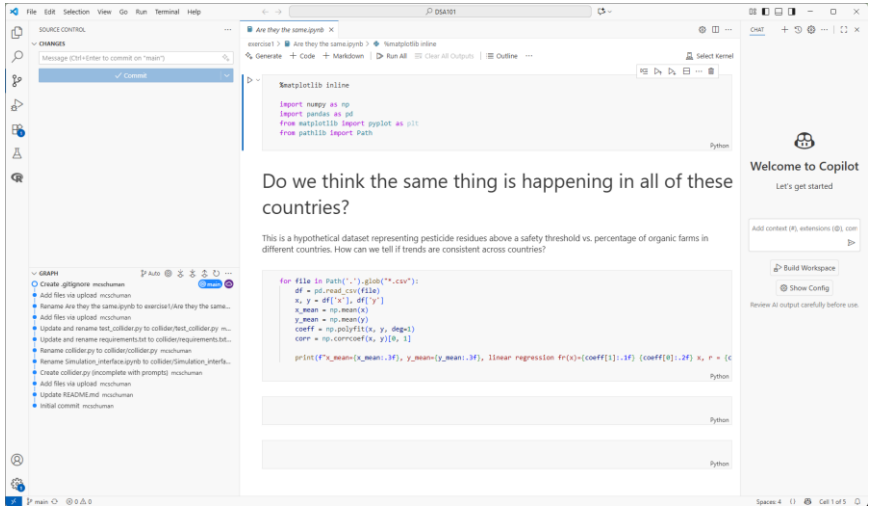
(1) [Make a GitHub account](https://github.com/signup)
if you don’t have one

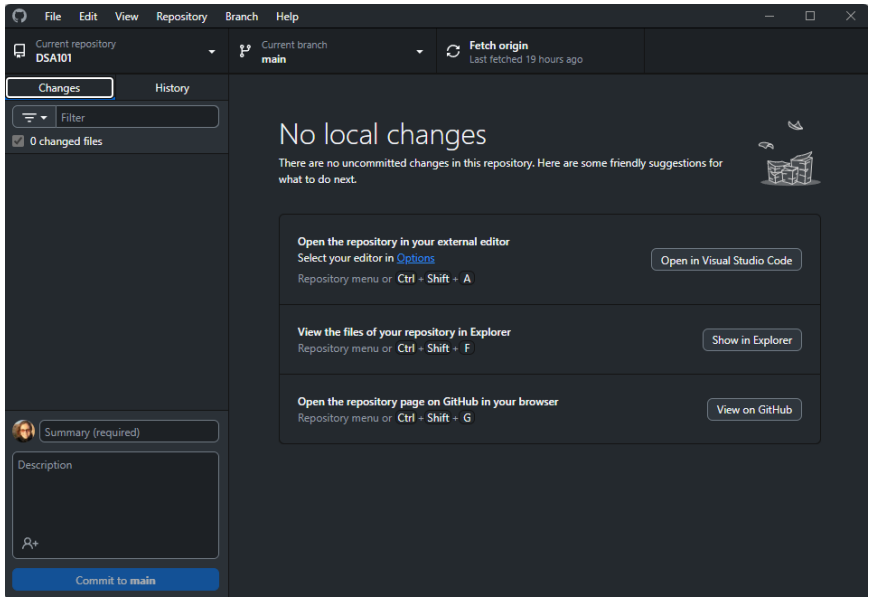


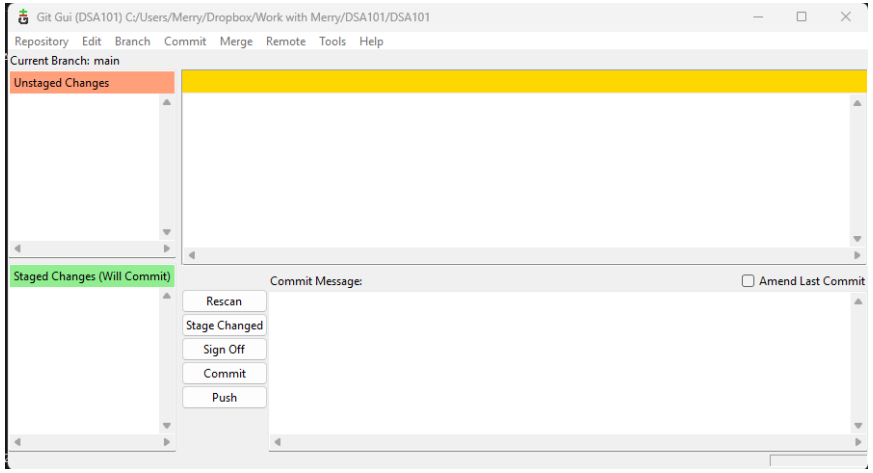
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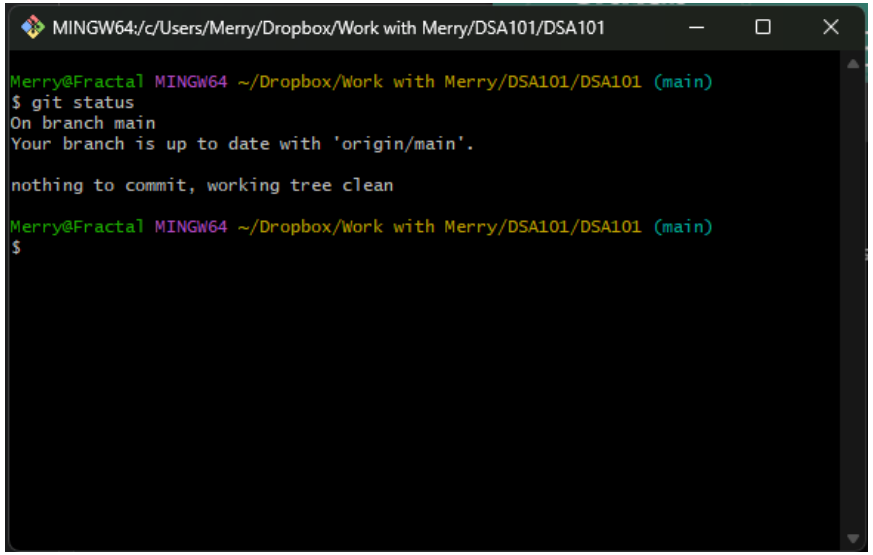
(1) Make a GitHub account if you don’t have one

(2) Install Git: there are many options!









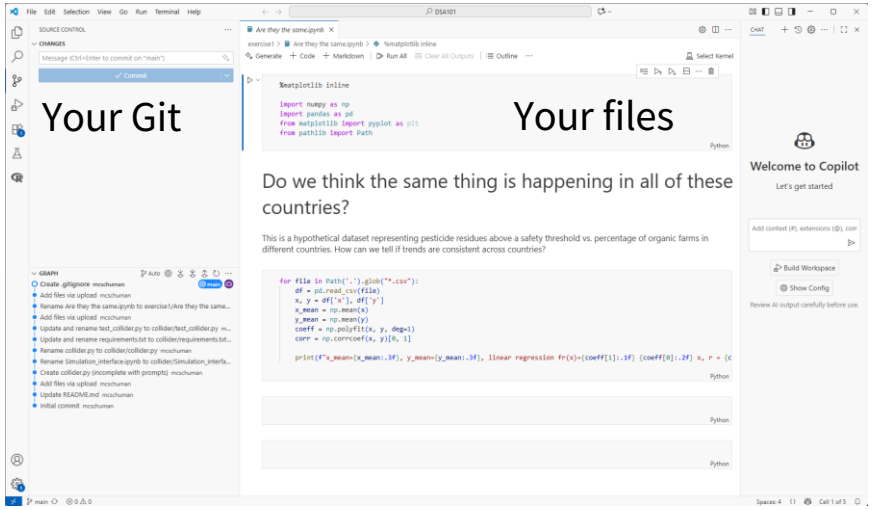
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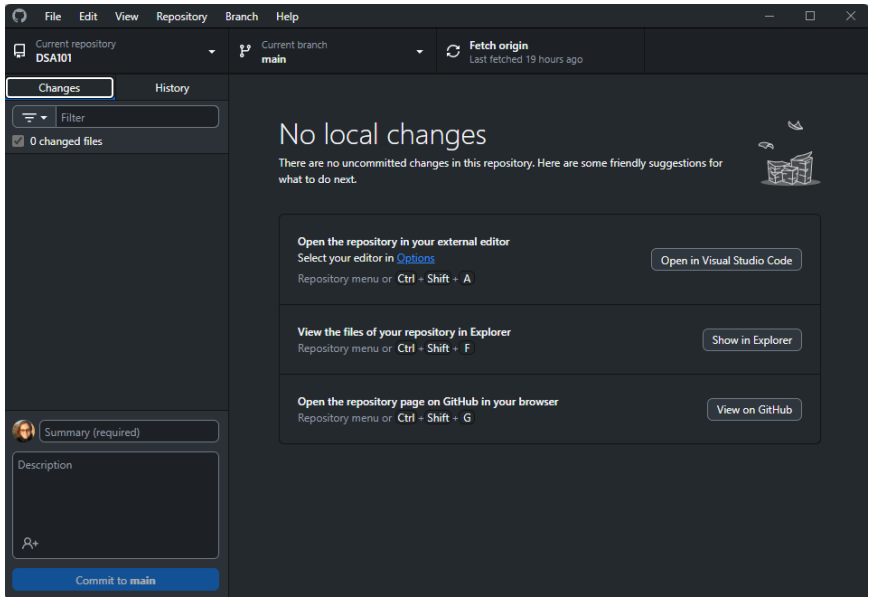
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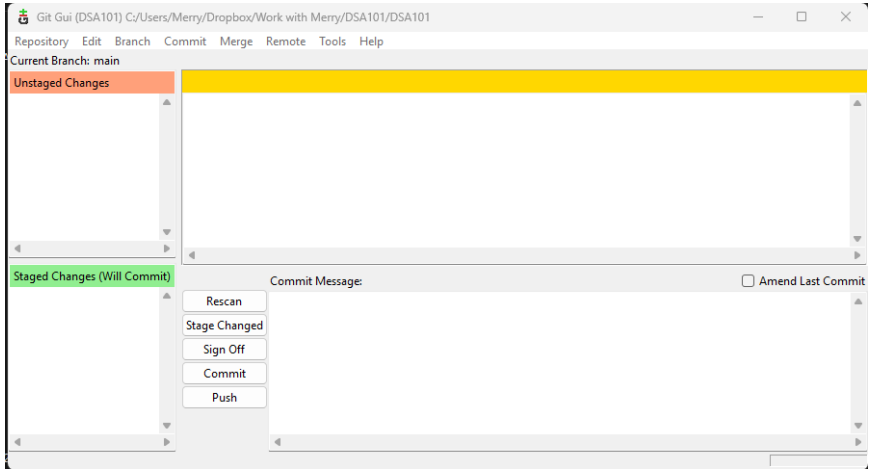
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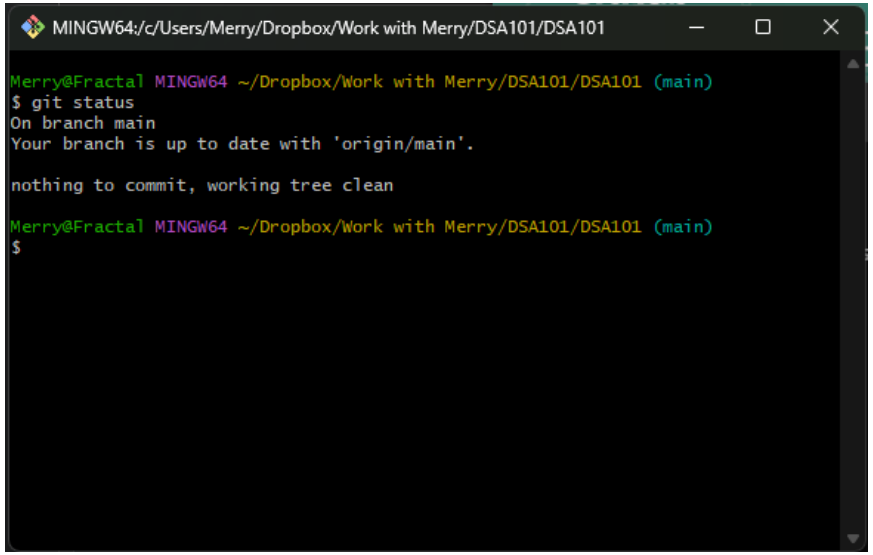
- **Easiest**: an IDE like Visual Studio Code: interface to interact with code and Git

Visual Studio Code or another IDE









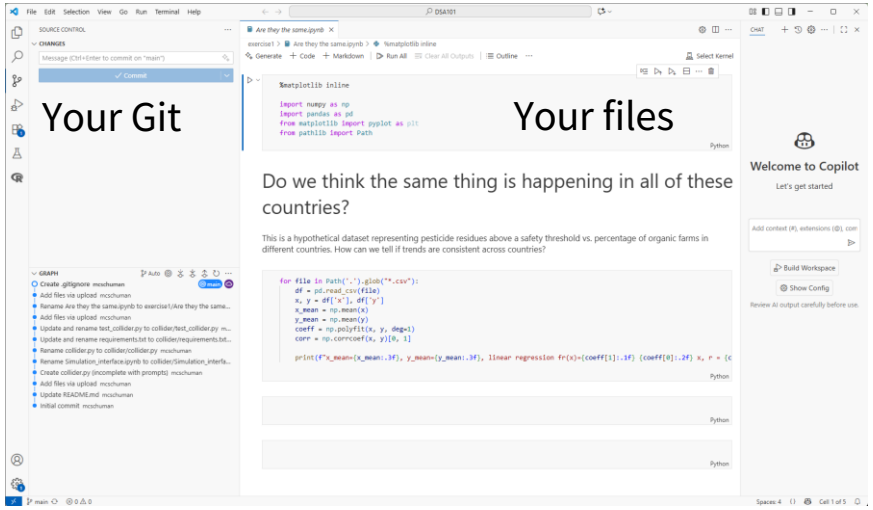
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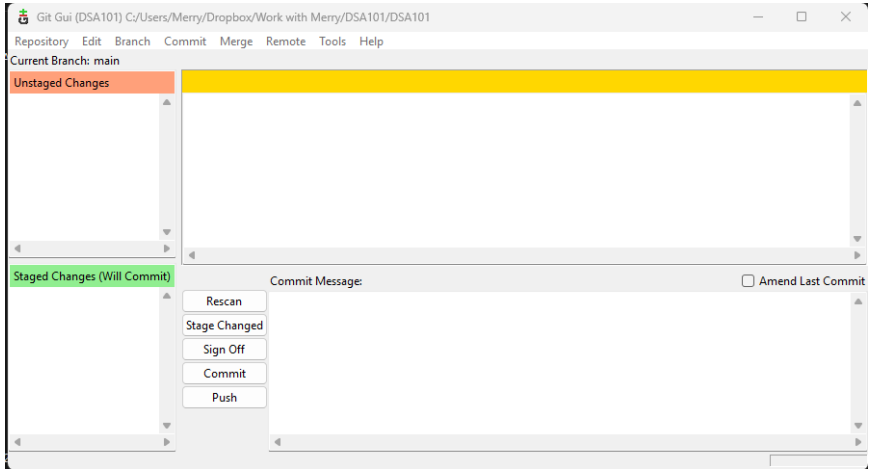
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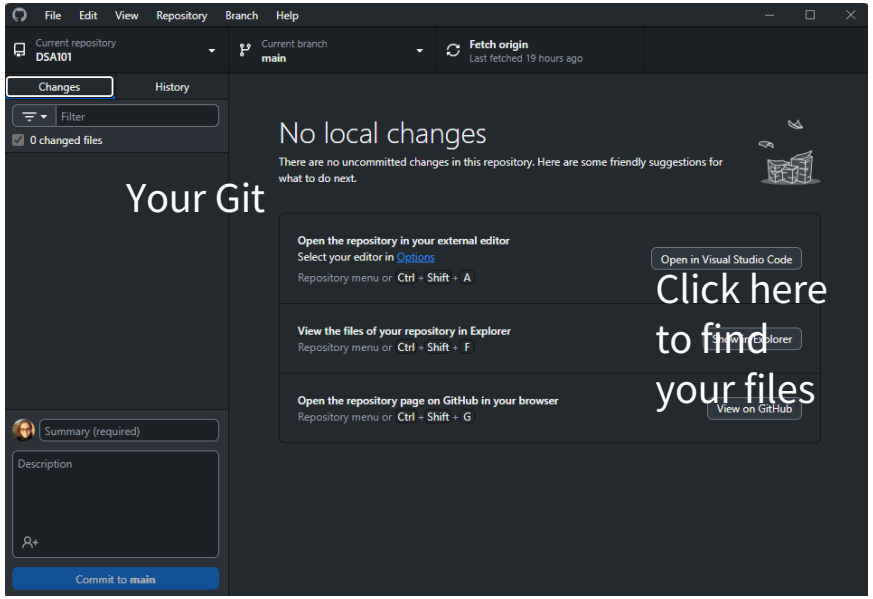
- **Easiest**: an IDE like Visual Studio Code: interface to interact with code and Git
- **Second easiest**: GitHub Desktop, a GUI that prompts you to use Git correctly

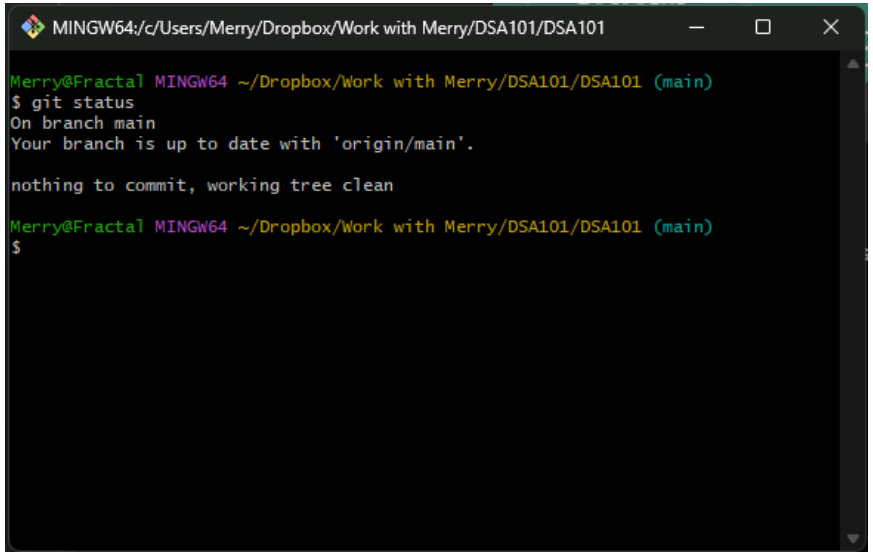
Visual Studio Code or another IDE





GitHub Desktop





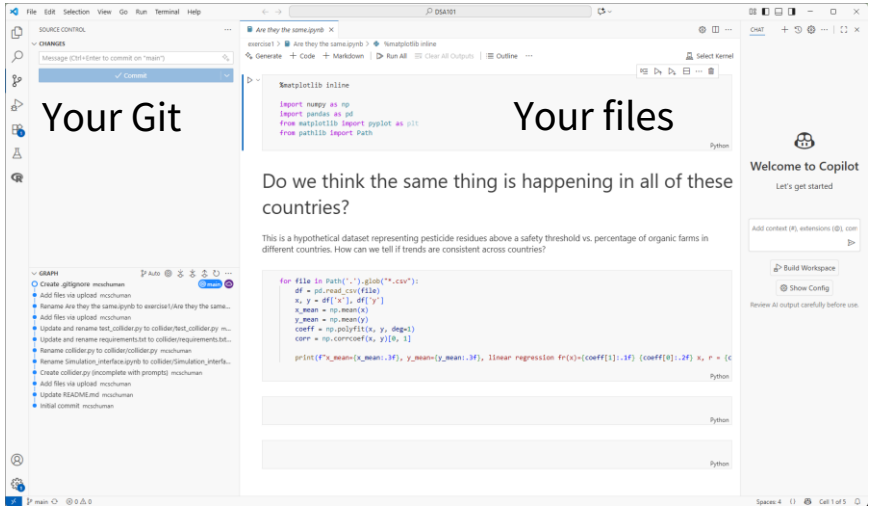
Let's start!

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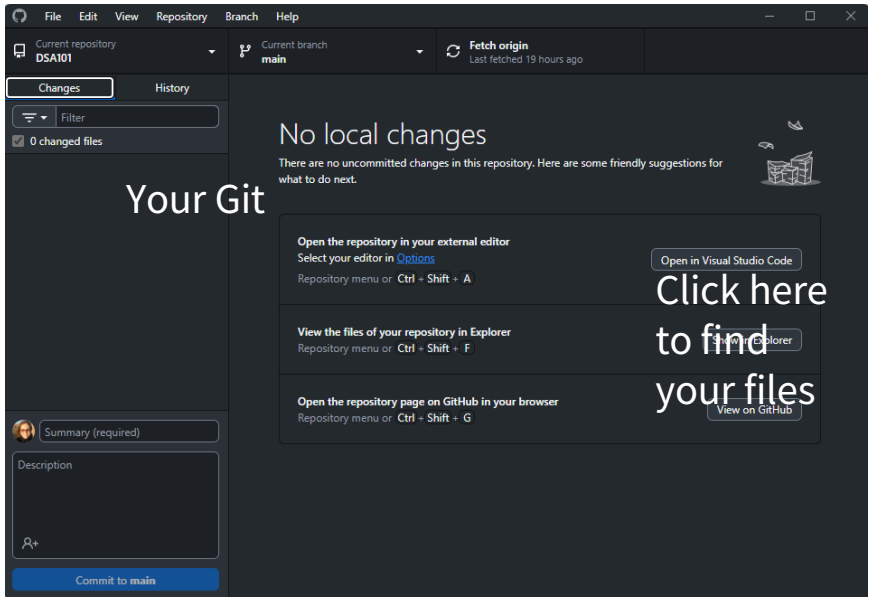
(2) [Install Git](#): there are many options!

- **Easiest**: an IDE like Visual Studio Code: interface to interact with code and Git
- **Second easiest**: GitHub Desktop, a GUI that prompts you to use Git correctly
- **Develop your computer skills (still fairly easy)**: install with Git installer, then interact with Git using Git GUI or Git Bash (or Terminal)

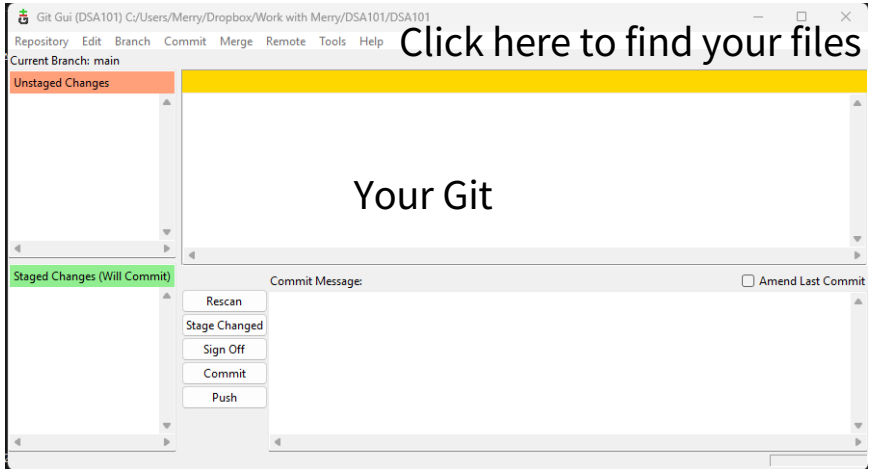
Visual Studio Code or another IDE



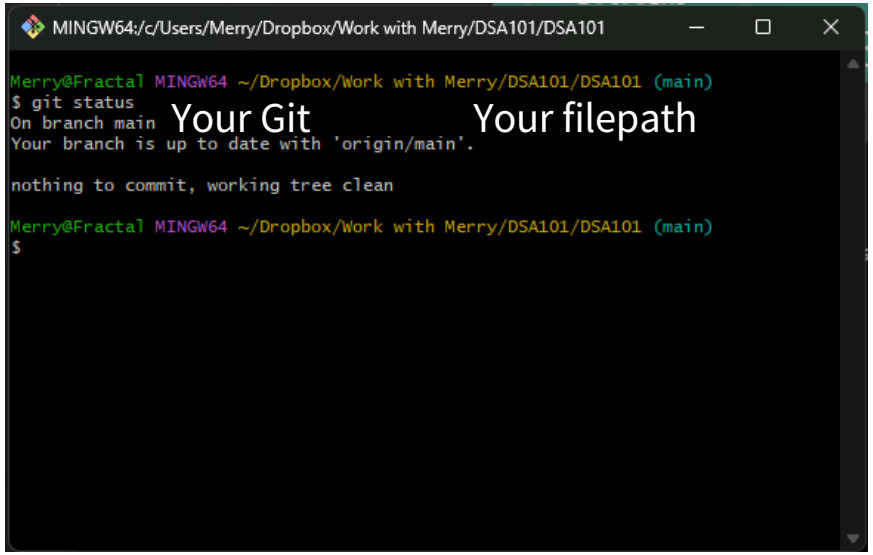
GitHub Desktop



Git GUI



Git Bash

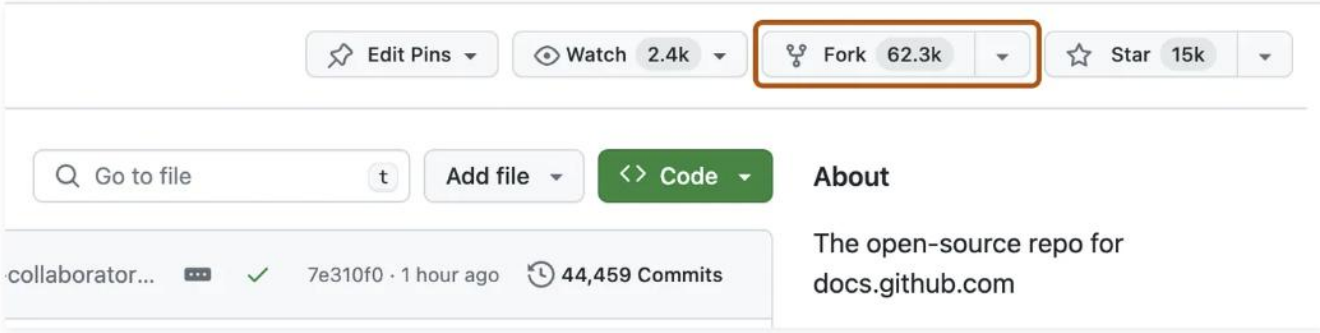


Let’s start!

(1) Make a GitHub account if you don’t have one

(2) Install Git: there are many options!

(3) Fork the DSA101_HS26 repository



3. Under "Owner," select the dropdown menu and click an owner for the forked repository.
4. By default, forks are named the same as their upstream repositories. Optionally, to further distinguish your fork, in the "Repository name" field, type a name.
5. Optionally, in the "Description" field, type a description of your fork.
6. Optionally, select **Copy the DEFAULT branch only**.

   For many forking scenarios, such as contributing to open-source projects, you only need to copy the default branch. If you do not select this option, all branches will be copied into the new fork.
7. Click **Create fork**.

https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/working-with-forks/fork-a-repo

Let’s start!

(1) Make a GitHub account if you don’t have one

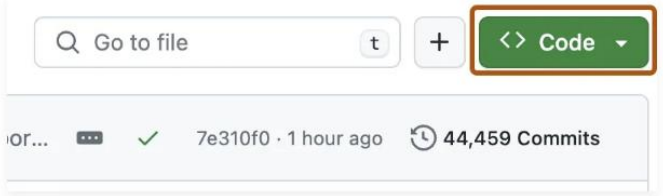
(2) Install Git: there are many options!

(3) Fork the DSA101 _HS26 repository

(4) Clone your fork

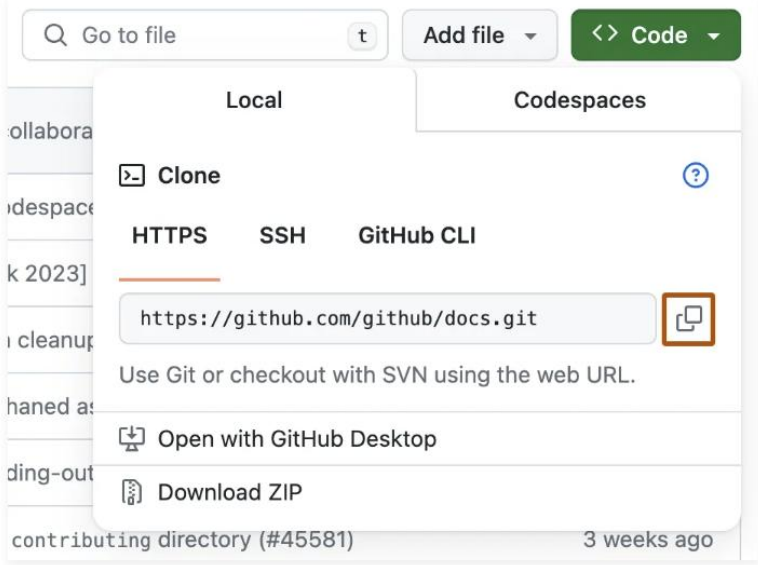
I expect you to understand how Git works at this level for the exam.

Explain process how to get repository



3 Copy the URL for the repository.

- To clone the repository using HTTPS, under "HTTPS", click.
- To clone the repository using an SSH key, including a certificate issued by your organization's SSH certificate authority, click **SSH**, then click.
- To clone a repository using GitHub CLI, click **GitHub CLI**, then click.



https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/working-with-forks/fork-a-repo

Next: Try exercise1

Watch GitHub Tutorials (ca 1h) and start with exercise 1

Are they the same?

We have a fictitious dataset from 12 different countries documenting the relationship between proportion of organic farms, and proportion of test sites in which pesticide residues are above a regulatory threshold.

You will characterize this relationship for each country and answer the question, is the relationship essentially the same across all countries in our dataset?

I expect you to understand how to look at data to answer specific questions for the exam.



https://4.bp.blogspot.com/-1gcbXO2ePh4/UY8Vzp5khqI/AAAAAAAAJbk/i6HUCVq43GE/s1600/pesticide+spraying+man.jpg